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A Whisker-like 2D Contact Sensor for Environmental Perception of Mobile Robots

Junjie Luo, Siyue Yao, Tao Yu, Yanjun Wang, Hao Wang and Genliang Chen

***Abstract*—Exploring the surrounding environment plays a critical role in robotic systems. This paper introduces a compact whisker-like sensor for planar contact point estimation based on the continuum mechanics of slender elastic beams. The proposed sensor achieves perception of contact point in the intermediate location of the beam by measuring several local deformations through a small number of strain gauges. An explicit model characterizes the force-deflection behavior of the sensor is developed, establishing the relationship between local bending and position of contact point under frictional and large-deformation conditions. A series of experiments are conducted to validate the accuracy of the sensor in detecting 2D contact points. Furthermore, the proposed sensors are integrated into a mobile robot, demonstrating the potential application in robotic environmental perception through width sensing, contour following and reconstruction.**

***Index Terms*—Contact Sensing Model, Sensor Based Control, Flexible Robotics, Environmental Perception.**

I. Introduction

In the natural world, animals have evolved diverse mechanisms to perceive their surrounding environment, including visual, auditory, olfactory, and tactile systems. Among these different sensing methods, contact sensing through whiskers plays a pivotal role for certain species, such as rodents and pinnipeds, enabling obstacle detection and navigation particularly under conditions where other sensory inputs are limited [1]–[4]. For robotic systems, the ability to perceive unknown environments is equally critical, where artificial whisker sensors have garnered significant interest due to their simple structural design and intuitive operational principles. Those sensors have been employed in different environment detecting and interacting applications [5]. However, challenges remain in practical use of whisker sensors due to limitations in sensing performance and manufacturing processes. Consequently, a notable gap persists between the capabilities of artificial whisker sensors and the sophisticated functionalities demonstrated by their biological counterparts [6].

Inspired by the tactile functions of biological whiskers, numerous artificial whisker sensors have been developed in recent years. These sensors have employed diverse transduction mechanisms, including piezoelectric [7], piezoresistive [8], capacitive [9], optical [10], magnetic [11] and strain-based mechanisms [12], to detect various environment features, such as texture, shape and wind speed. In addition to measuring a single feature, some whisker sensors incorporate sensing units based on different principles within a single sensor, enabling the detection of multiple environmental properties. Harada *et al.* developed an e-whisker that integrates strain and temperature sensors, allowing simultaneous detection of object temperature and strain [13]. A multimodal whisker sensor was designed by Wegiriya *et al.*, which employs piezoelectric and Hall sensors to measure high-frequency and low-frequency vibration signals of the whisker [14].

Existing whisker sensors can be categorized into two types: passive sensors, where the whisker is fixed and responds to external stimuli, and active sensors, which use sweeping motions or similar movements. An *et al.* developed a bendable whisker mechanoreceptor and integrated it into a mobile robot, enabling obstacle recognition, object surface scanning and reconstruction through the robot's movements [15]. Yu *et al.* developed an active whisker sensor actuated by electromagnetic coils for distance, shape, size, and orientation sensing [16]. Active perception expands the sensing range and improves efficiency, however, it imposes higher demands on the material properties of the whiskers and the characteristics of the sensing units.

Utilizing the developed whisker sensors, various environmental perception and interaction applications have been demonstrated in different robotic systems, including object texture perception [17], contour reconstruction [18], obstacle avoidance [19], and flow measurement [20].

Despite these advancements, current whisker sensors face several limitations. On the one hand, the fabrication processes of some sensors are complex, including the integration of delicate electronic components, precise alignment, and specialized materials, which leads to increased production costs and limits their suitability for practical robotic applications. On the other hand, while the whisker sensors of creatures can detect contacts in the middle section of the whisker, some whisker sensors can only detect contacts occurring at the whisker tips. Although certain sensors have achieved the estimation of

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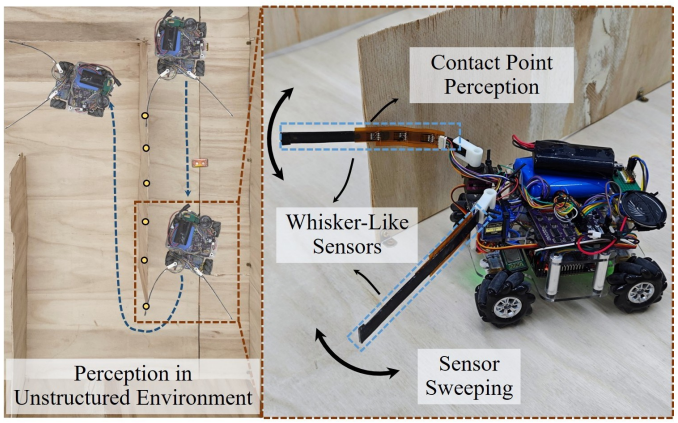


Fig. 1. The Developed Whisker-Like Sensors for Unknown Environment Perception.

contact point in the middle section of the whisker shaft [21]–[23], these approaches either necessitate bulky force/torque measurement and sampling devices, reducing the system's portability, or rely on whisker models that are only applicable under frictionless contact conditions. Low computational efficiency is another limitation for whisker sensors, resulting in the difficulty to meet the demands of real-time applications. These challenges highlight the need for whisker sensors that are simple in fabrication, low in cost, accurate in contact point sensing, and efficient in computation.

Elastic structures have found extensive applications in robotics owing to their compliant and low-hysteresis characteristics [24]–[26]. Based on the continuum mechanics of slender elastic beams, a whisker-like sensor is proposed in this paper. As shown in Fig. 1, unlike most existing whisker sensors, the proposed sensor introduces several measurement points on the beam surface, where only a small number of strain gauges is required for local bending measurements. With an explicit model that characterizes the force-deflection behavior, position of friction-inclusive external contact acting in the intermediate location of the sensor can be accurately estimated, even under large-scale deformations. The use of industrial strain gauges and elastic material endows the sensor with an integrated structure and a low-hysteresis response. Owning to its straightforward structure, an actuator can be readily mounted at one end of the beam, thus achieving active sensing. Demonstrations are provided by attaching two of the sensors on a mobile robot, which exhibits the potential applications in robotic systems for environmental perception based on the contact point sensing ability of the proposed sensors.

The rest of this paper is arranged as follows: Section II introduces the modeling process of the proposed sensor. Sensor fabrication and calibration are provided in Section III. Section IV validates the contact sensing performance of the sensor through different experiments. Demonstrations of environmental perception and reconstruction are exhibited through a mobile robot in Section V. Finally, conclusions and future works are presented in Section VI.

II. Modeling of the Whisker-Like Sensor

In this section, kinetostatics modeling of the elastic beams which are used as contact sensors are presented, followed by a sensing model that reconstructs the external contact conditions utilizing local bending information.

A. Modeling of Elastic Beams Under External Contact

Based on our prior work [27], elastic links can be discretized into a number of small segments. For each segment, a spatial six-DOF linkage with rigid bodies and passive elastic joints can be obtained [28], which can approximate the force deflection behavior of the segments. By connecting all the segments, a hyper-redundant rigid-body mechanism with elastic joints can be constructed. For the proposed sensor, which is constrained to planar motion in its operational conditions, since the thickness of the elastic beam is much smaller than its width, the effects of shearing and compression are neglectable comparing to the bending one. As a result, for the discrete elements in the subsequent modeling framework, only the single-DOF bending related revolute joints will be considered.

As shown in Fig. 2, when external contact of single point happens, a spatial frame $\{\mathrm{S}\}$ and a floating contact frame $\{\mathrm{C}\}$ are constructed at the base and the contact point of the beam. The origin of the contact frame $\{\mathrm{C}\}$ is specified through the curved length l_c from the sensor base. Using product-of-exponential (POE) formulation for forward kinematics, pose of the contact frame $\{\mathrm{C}\}$ with respect to $\{\mathrm{S}\}$ can be represented as

$$\mathbf{g}_{sc}(\boldsymbol{\theta}) = \exp(\hat{\boldsymbol{\zeta}}_1\theta_1)\cdots\exp(\hat{\boldsymbol{\zeta}}_{n_c}\theta_{n_c})\,\mathbf{g}_{sc,0} \tag{1}$$

where $\hat{\boldsymbol{\zeta}}_i \in se(3)$, $i = 1, \cdots, n$ denote the twists of the joints in home configuration. $\boldsymbol{\theta} = [\theta_1, \cdots, \theta_n] \in \mathbb{R}^{n\times 1}$ denotes the bending related joint displacement of the whole link, where n is the total number of joints, and n_c is the joint number that contact happens. $\mathbf{g}_{sc,0} = \mathbf{g}_{sc}(0)$ relates to the initial pose of $\{\mathrm{C}\}$. The Jacobian matrix $\mathbf{J_c}$ associated with $\{\mathrm{C}\}$ can be derived as

$$\left(\dot{\mathbf{g}}_{sc}\mathbf{g}_{sc}^{-1}\right)^{\vee} = \mathbf{J_c}\dot{\boldsymbol{\theta}} \Rightarrow \mathbf{J_c} = [\boldsymbol{\xi}_1, \cdots, \boldsymbol{\xi}_{n_c}, \boldsymbol{0}, \cdots, \boldsymbol{0}] \in \mathbb{R}^{6\times n} \tag{2}$$

Here $(\cdot)^{\vee}$ is the 'vee' operator which maps an $se(3)$ element in 4×4 standard form to 6×1 vector form. $\boldsymbol{\xi}_i = \mathrm{Ad}(\exp(\hat{\boldsymbol{\zeta}}_1\theta_1)\cdots\exp(\hat{\boldsymbol{\zeta}}_{i-1}\theta_{i-1}))\boldsymbol{\zeta}_i$ corresponds to the joint twists in the current configuration with $\mathrm{Ad}(\cdot)$ being the adjoint transformation of $SE(3)$ elements.

Taking the contact force into consideration, the static equilibrium condition of the joints should be satisfied, which can be written as

$$\boldsymbol{\tau} = \mathbf{K}_{\boldsymbol{\theta}}\boldsymbol{\theta} - \mathbf{J}_c{}^T\boldsymbol{F}_c = \boldsymbol{0} \tag{3}$$

where $\mathbf{K}_{\boldsymbol{\theta}} = \mathrm{diag}(1/c_1, \cdots, 1/c_n)$ is the diagonal matrix of joint bending stiffness. And $\boldsymbol{F}_c \in \mathbb{R}^{6\times 1}$ represents the external contact force. The Jacobian matrix $\mathbf{J}_c$ maps $\boldsymbol{F}_c$ to the joint reactions through joint twists $\boldsymbol{\xi}_i$ in the current configuration.

The proposed model incorporates contact involving static friction, taking into account both the normal compress force $\boldsymbol{F}_n$ and the shear force $\boldsymbol{F}_s$ during the modeling process. Thus the contact force is modeled to be a combination of its two

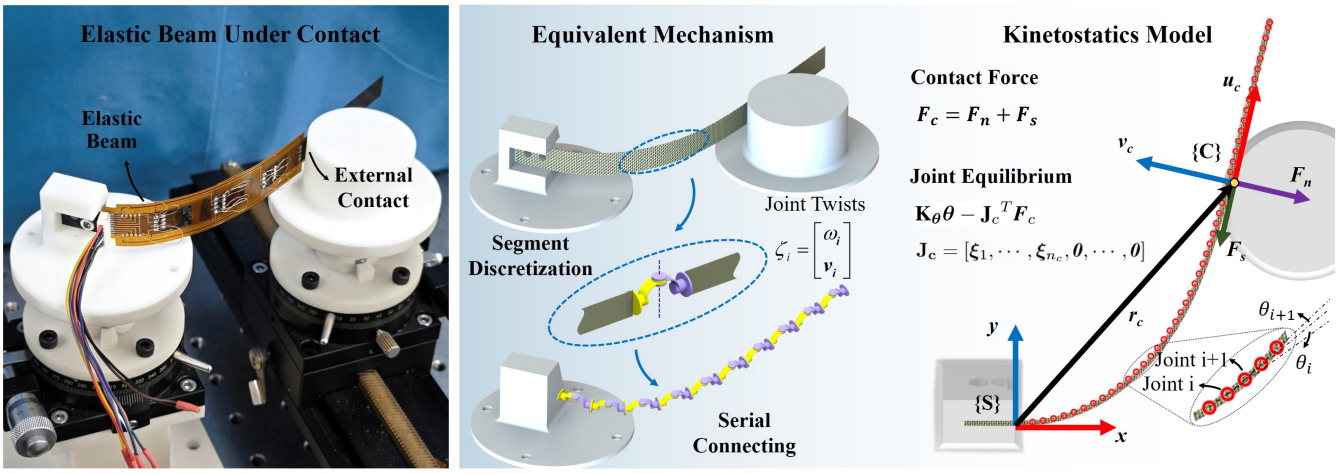


Fig. 2. Equivalent Mechanism and Kinetostatics Model of Elastic Beam Under Single-Point Contact.

components as $\boldsymbol{F}_c = \boldsymbol{F}_s + \boldsymbol{F}_n$. The normal component at the origin of $\{\mathrm{C}\}$ can be represented as

$$\boldsymbol{F}_n = f_n \begin{bmatrix} \hat{\mathbf{r}}_c \boldsymbol{v}_c \\ \boldsymbol{v}_c \end{bmatrix} = f_n \boldsymbol{W}_n \tag{4}$$

where f_n is the magnitude of the normal compress force. $\boldsymbol{W}_n \in \mathbb{R}^{6\times 1}$ corresponds to the unit wrench with $\boldsymbol{r}_c \in \mathbb{R}^{3\times 1}$ being the contact position, $\boldsymbol{v}_c \in \mathbb{R}^{3\times 1}$ being the unit direction vector of $\boldsymbol{F}_n$, which is normal to the deformed sensor at $\{\mathrm{C}\}$. Since static friction is taken into account, the shear component can be modeled as an independent variable applied tangential to the deformed beam at the contact point, which satisfies $\boldsymbol{F}_s = f_s \boldsymbol{W}_s$, with f_s being the shear force magnitude and $\boldsymbol{W}_s$ being the unit wrench of the shear component.

Considering that the geometric constraints and joint balance constraints need to be satisfied simultaneously under the equilibrium state, the kinetostastics model of the elastic beam under external single-point contact can be expressed in the form of nonlinear equations as

$$\boldsymbol{C}(\boldsymbol{\theta}, \boldsymbol{F}_c) = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} \left(\ln(\mathbf{g}_{sc}\, \mathbf{g}_{tc}^{-1})\right)^{\vee} \\ \mathbf{K}_{\boldsymbol{\theta}}\boldsymbol{\theta} - \mathbf{J}_c{}^T \boldsymbol{F}_c \end{bmatrix} = \boldsymbol{0} \tag{5}$$

where $\boldsymbol{y}$ represents the pose deviation of $\{\mathrm{C}\}$ from its target $\mathbf{g}_{tc}$. The model corresponds to $n+6$ equations and $n+6$ variables, and is thus solvable through an analytical way.

The system Jacobian matrix of (5) can be derived as

$$\nabla_{\mathbf{C}} = \begin{bmatrix} \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{\theta}} & \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{F}_c} \\ \frac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{\theta}} & \frac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{F}_c} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_c & \mathbf{0} \\ \mathbf{K}_{\boldsymbol{\theta}} - \mathbf{K}_{\mathbf{J}_c} & -\mathbf{J}_c^T \end{bmatrix} \tag{6}$$

where $\mathbf{K}_{\mathbf{J}_c}$ is a configuration- dependent item of the overall joint stiffness. Please refer to [27] for more details.

Then, following the update theme of Newton-Raphson method, the equilibrium configuration of the elastic beam can be identified as

$$\begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{F}_c \end{bmatrix}_{k+1} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{F}_c \end{bmatrix}_{k} - (\boldsymbol{\nabla}_{\boldsymbol{C}})_k^{-1} \boldsymbol{C}_k \tag{7}$$

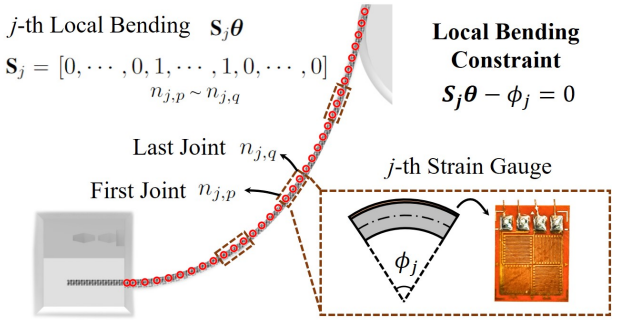


Fig. 3. Local Bending Measured by Strain Gauges Attached at Distinct Sections.

where k is the kth step of the iterative searching algorithm.

By employing the kinetostatics contact model, it is feasible to efficiently determine the displacement of each equivalent joint $\boldsymbol{\theta}$ in the elastic beam under specified external contact conditions, thereby ascertaining localized deformation. Consequently, this model is well-suited for sensor calibration and the generation of extensive deformation-contact data.

B. Modeling of Contact Point Perception

As shown in Fig. 3, in order to sense the exact position of external contact, m sections are selected along the elastic beam, whose measured local deformation should equal to the local deformation computed from joint displacement $\boldsymbol{\theta}$. Those extra constraints are expressed as

$$\mathbf{P} = \mathbf{S}_\theta \boldsymbol{\theta} - \boldsymbol{\phi}_s = \boldsymbol{0} \tag{8}$$

Here $\mathbf{S}_\theta$ is a constant selection matrix indicating the position of each deflection angle measurement section such that

$$\mathbf{S}_\theta = [\mathbf{S}_1, \cdots, \mathbf{S}_j, \cdots, \mathbf{S}_m]^T \in \mathbb{R}^{m\times n} \tag{9}$$

where $\mathbf{S}_j = [0, \cdots, 0, 1, \cdots, 1, 0, \cdots, 0]$ with its $n_{j,p}$ to $n_{j,q}$ elements being 1 and other elements being 0. $n_{j,p}$ and $n_{j,q}$ indicate the first and last joint covered by each local measurement area. Since each local measurement section is

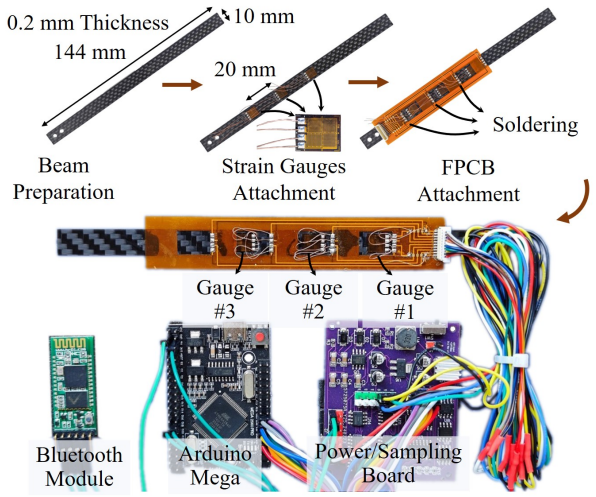


Fig. 4. Fabrication Process of the Whisker-Like Sensor.

distributed at different positions along the beam and includes distinct joints, the positions of 1 in each row of the selection matrix are unique. This characteristic ensures that the rows of the selection matrix $\mathbf{S}_\theta$ are always linearly independent. $\boldsymbol{\phi_s} = [\phi_1, \cdots, \phi_j, \cdots, \phi_m]^T \in \mathbb{R}^{m\times 1}$ is the measured deflection angles of the m sections through the strain gauges covering the same joints indicated by $\mathbf{S}_\theta$.

Combining the static equilibrium (3) and the local deformation constraints (8), the sensing model of the elastic beam can be represented in the general form of nonlinear algebraic equations as

$$\boldsymbol{S}(\boldsymbol{\theta},\, f_n,\, f_s,\, l_c) = \begin{bmatrix} \mathbf{P} \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} \mathbf{S}_\theta \boldsymbol{\theta} - \boldsymbol{\phi}_s \\ \mathbf{K}_{\boldsymbol{\theta}} \boldsymbol{\theta} - {\mathbf{J}_c}^T \boldsymbol{F}_c \end{bmatrix} = \boldsymbol{0} \tag{10}$$

As indicated by (10), the nonlinear algebraic equations contains $n+m$ equations with $n+3$ unknowns, a least number of $m = 3$ is required to solve the equations. In this paper, 3 sections are used for experiment. When friction force is neglected, $\boldsymbol{F}_c$ reduces to $\boldsymbol{F}_n$, and the number of unknown variables decreases to $n+2$, which makes (10) a set of over-constrained equations, so that the least square solution can be acquired as the best estimation in that case.

To solve the equations, the corresponding system Jocobian of $\boldsymbol{S}$ can be derived as

$$\nabla_{\mathbf{S}} = \begin{bmatrix} \mathbf{S}_\theta & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{K}_{\boldsymbol{\theta}} - \mathbf{K}_{\mathbf{J}_c} & -\mathbf{J}_c^T \boldsymbol{W}_n & -\mathbf{J}_c^T \boldsymbol{W}_s & \mathbf{J}_{\boldsymbol{\tau},l} \end{bmatrix} \tag{11}$$

Here, the partial derivative of $\boldsymbol{\tau}$ with respect to the contact location l_c can be obtained as

$$\mathbf{J}_{\boldsymbol{\tau},l} = \frac{\partial \boldsymbol{\tau}}{\partial l_c} = -f_n \mathbf{J}_c^T \frac{\partial \boldsymbol{W}_n}{\partial l_c} - f_s \mathbf{J}_c^T \frac{\partial \boldsymbol{W}_s}{\partial l_c} \tag{12}$$

with

$$\frac{\partial \boldsymbol{W}_n}{\partial l_c} = \begin{bmatrix} \hat{\mathbf{r}}_c \boldsymbol{\Psi}_c - \hat{\mathbf{v}}_c \boldsymbol{\Gamma}_c \\ \boldsymbol{\Psi}_c \end{bmatrix} [\boldsymbol{0},\, \boldsymbol{u}_c]_{6\times 2} \tag{13}$$

where $\boldsymbol{\Psi}_c = [-\hat{\mathbf{v}}_c, \mathbf{0}_3] \in \mathbb{R}^{3\times 6}$ and $\boldsymbol{\Gamma}_c = [-\hat{\mathbf{r}}_c, \mathbf{I}_3] \in \mathbb{R}^{3\times 6}$. The shear component can be derived similarly.

The equilibrium configuration of the elastic beam can be then identified following the similar update theme as (7). Here, the Moore-Penrose inverse of $\boldsymbol{\nabla_S}$ is used since system Jacobian may not be a square matrix with different numbers of selected sections.

As the curved length l_c is calculated from the equilibrium configuration, the joint number where contact happens is determined. As a result, the external contact position can be estimated from (1).

III. Sensor Fabrication and Calibration

In this section, a whisker sensor prototype is constructed. As shown in Fig. 4, the sensor utilizes a $10mm \times 0.2mm \times 144mm$ carbon fiber plate as its structural element, providing sufficient compliance for significant deformation under external loading conditions. The fabrication process of the sensor is relatively straightforward. Initially, three full-bridge strain gauges are adhered to the elastic plate at intervals of $20\,mm$, and the local deformation of these three sections under bending state can be measured. Subsequently, a flexible printed circuit board (FPCB) is affixed to the surface of the elastic plate. Finally, the four lead wires of each strain gauge are soldered to the designated pads on the flexible circuit board. Through the integrated flexible circuit board, the three strain gauges can be easily powered, and the output voltage signal of which can be guided to the sampling module in a compact and convenient way. The voltage signals are obtained through an AD7895 converter and sent to a laptop through Bluetooth connection. A sampling frequency of more than $100\,Hz$ can be achieved when six strain gauges are connected to the system. The entire sensor prototype system is powered by $7.4\,V$ lithium battery, enabling portable and wireless operation.

The developed sensor prototype underwent rigorous reliability testing through more than 10,000 cyclic loading repetitions, demonstrating minimal fatigue effects and plastic deformation, and low-hysteresis characteristics, thereby confirming its robust operational durability. Detailed test results are presented in the supplementary material.

Calibration plays a pivotal role in enhancing the accuracy of robotic systems [29], [30]. For the developed sensor prototype, a sensor calibration experiment is required to establish the relationship between the three strain gauge voltage readings and their corresponding deformations. As shown in Fig. 5, an experimental setup is built so that one end of the whisker sensor is mounted on a fixed base, and a slender object is fixed on a slider that could translate and rotate in a plane to apply external contact to the whisker-like sensor. A freely rotating pulley is mounted at the tip of the slender object. Consequently, during the calibration process, the contact between the object and the elastic beam is assumed to be frictionless. The pulley center of the slender object is moved to 54 distinct positions on both sides of the sensor for calibration, which are marked by dots in Fig. 5, where the according bending angles of the strain gauges are solved through (6) and (7). At meantime, the voltage readings of the strain sensor are recorded for each configuration. A calibration configuration is also presented in Fig. 5, demonstrating close agreement

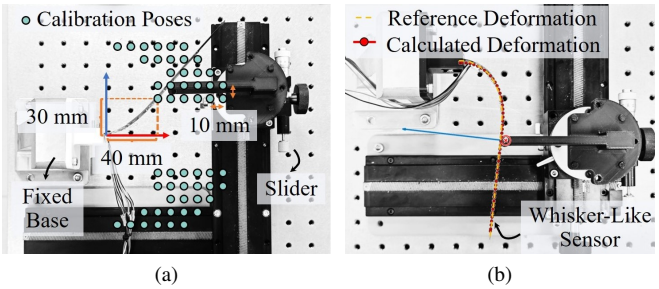


Fig. 5. Calibration of the Proposed Sensor: (a) Calibration Setup. (b) Visualization of a Specific Configuration.

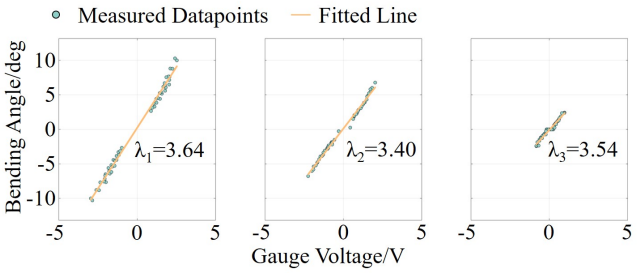


Fig. 6. Identification of the Proportional Coefficients for the Three Strain Gauges.

between the model calculated beam deformation and the physically captured counterpart through camera during the calibration process.

Previous research [31]–[33] related to material mechanics and strain measurements has demonstrated a linear relationship between the local bending and the voltage of the strain gauges as

$$\phi_j = \frac{8l_0}{Kh_0U_0} U_{s,j} = \lambda_j U_{s,j} \tag{14}$$

where U_0 and $U_{s,j}$ denote the supply and measured voltages of the jth strain gauge, K is a constant sensitivity ratio related to the wire material. l_0 and h_0 are the effective length of the gauges and the thickness of the beam.

The relationship between strain gauge deformations and voltage outputs is established through linear regression analysis. Fitting results for different gauges are presented in Fig 6, where the proportional coefficients λ_j are obtained through linear regression. Under the obtained calibration parameters, the sensed contact positions estimated from (10) are compared with the reference positions across the 54 data sets, resulting in an average calibration residual of $3.56\,mm$.

IV. Experimental Validation

In this section, experiments are conducted to verify the accuracy of the proposed sensor. Contact point perception ability in the intermediate location of the senor is examined in a passive manner, and the contact points obtained in an active contact state are used to identify parameters of cylindrical objects.

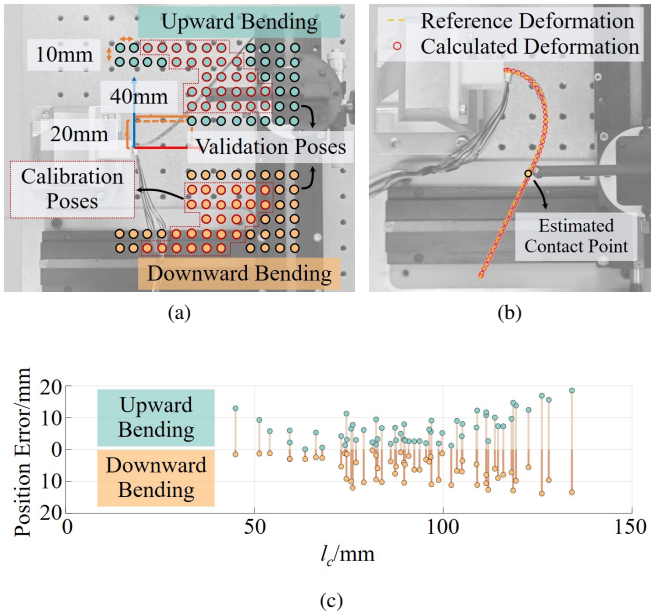


Fig. 7. Experimental Validation of Contact Point Perception: (a) Bidirectional Validation Poses. (b) Visualization of a Specific Configuration. (c) Distribution of Position Error for Contact Points.

A. Validation of Contact Point Perception

To verify the contact point sensing ability of the proposed sensor, 112 groups of validation poses covering both bending directions are selected, as shown in Fig. 7. The range of validation configurations slightly extended beyond the calibration ones, encompassing contact points closer to both the fixed end and the tip of the sensor, which covers more than 60% of the total sensor length. It is noteworthy that during the verification process, the tip pulley of the contact object is fixed with bolts, necessitating the use of the sensing model that accounts for frictional force.

For all 112 verification data sets, the average position error between the sensed contact points and the reference positions is $6.45\,mm$ (4.45% of the total beam length). The distribution of position error at different location of the sensor is illustrated in Fig. 7, where the top and bottom half of the data indicates the bending direction of the sensor. The results demonstrate that the sensing errors are generally consistent across different locations on both bending directions. When focusing on the $50\,mm$ to $100\,mm$ contact range from the fixed base (similar range to that of calibration), the sensor achieves enhanced accuracy with a reduced mean error of $4.78\,mm$ (3.30% of the total beam length). A representative case is included in Fig. 7, where the sensor tip is pushed backward from its initial direction, and the resulted strain gauge bending angle reaches more than $10°$. This substantial deformation effectively demonstrates the sensor's capability to accurately estimate contact points even under challenging conditions involving frictional force and large-scale deformations.

B. Object Parameter Estimation

In addition to passively sensing external contact in a stationary state, the proposed sensor can obtain more environmental

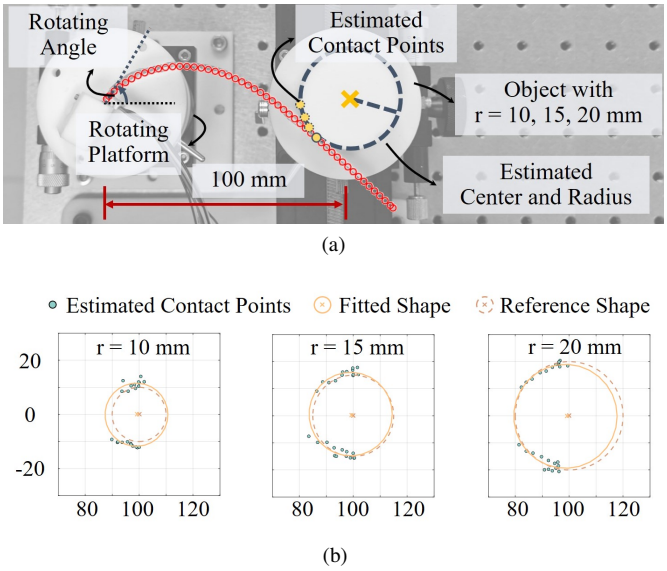


Fig. 8. Object Parameter Estimation: (a) Experimental Setup. (b) Estimated Contact Positions and Fitted Circular Shape for Different Objects.

information through active movement. To verify this feature, one end of the sensor is mounted on a platform that can rotate in a plane, and a cylindrical object is placed near the sensor, with its center kept at a fixed distance of 100 mm from the beam base, as shown in Fig. 8. A contact sensing example for a cylindrical object of $20\,mm$ radius with a 50° rotation angle is provided in Fig. 8, where the estimated beam deformation and the contact position coincides well with the real photo.

When the end of the beam rotates actively, the sensor sequentially contacts multiple points along the obstacle's circumference, generating a discrete set of contact coordinates $(x_j, y_j), j = 1, \cdots, p$ in the global reference frame, where p is the total number of contact points. The circular shape of the obstacle is then reconstructed by fitting each contact point using the least squares method. The center coordinate of the object (x_o, y_o) and its radius r_o can be identified as

$$\begin{bmatrix} x_o \\ y_o \\ {x_o}^2 + {y_o}^2 - {r_o}^2 \end{bmatrix} = A^{-1}\mathbf{b} \tag{15}$$

where

$$A = \begin{bmatrix} \sum_{j=1}^{p} x_j & \sum_{j=1}^{p} y_j & p \\ \sum_{j=1}^{p} x_j y_j & \sum_{j=1}^{p} y_j^2 & \sum_{j=1}^{p} y_j \\ \sum_{j=1}^{p} x_j & \sum_{j=1}^{p} y_j & \sum_{j=1}^{p} 1 \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} \sum_{j=1}^{p} (x_j^2 + y_j^2) x_j \\ \sum_{j=1}^{p} (x_j^2 + y_j^2) y_j \\ \sum_{j=1}^{p} (x_j^2 + y_j^2) \end{bmatrix}$$

In the experiment, identification is performed on circular objects of three different sizes with radii of 10, 15, and $20\,mm$, with data recorded at a rotating angle interval of 5 degrees. Fig 8 shows the perception result of the different objects. The radius and center position estimation errors of each object are also summarized in Table I, with an average radius error of $0.97\,mm$, and an average center position error of $1.17\,mm$.

TABLE I
OBJECT PARAMETER IDENTIFICATION RESULT AND ERROR

	Object Parameters	Reference	Identified	Error
#1	Radius/mm	10.00	11.66	1.66
	Center Position/mm	(100.00, 0.00)	(99.02, -0.10)	0.98
#2	Radius/mm	15.00	15.42	0.42
	Center Position/mm	(100.00, 0.00)	(99.05, 0.63)	1.14
#3	Radius/mm	20.00	19.18	-0.82
	Center Position/mm	(100.00, 0.00)	(98.63, -0.19)	1.39

C. Discussion on Position Error of Contact Point Perception

The position estimation accuracy of the proposed sensor system is primarily affected by several error sources. First, the current prototype employs exactly three strain gauges, which constitutes the minimum required for solving the nonlinear equations (10), resulting in heightened sensitivity to individual strain measurement variations. Consequently, noise in the measurement and signal acquisition circuit results in deformation measurement error, ultimately compromising contact point estimation accuracy. Second, fabrication tolerances in 3D-printed structural components (including the sensor mounting base and contact objects) introduce discrepancies between actual contact point and theoretical assumptions. These dimensional variations may additionally induce out-of-plane deformations and shear forces that violate model constraints. Furthermore, the present design permits object contact directly with strain gauge solder joints or lead wires when approached from the strain gauge attaching side, creating a mismatch with the modeled assumption of contact on elastic beam surface, thereby generating estimating errors.

To address these limitations, future implementations could incorporate additional strain gauges coupled with enhanced signal conditioning circuitry to improve measurement robustness, while transitioning to precision-machined components would better enforce planar motion constraints. Improvements could also be achieved through advanced fabrication techniques such as direct deposition of strain-sensitive resistance grids on FPCB, eliminating soldered connections and achieving higher integration. These combined enhancements would address the current design's vulnerabilities while maintaining the system's fundamental operational principles.

V. DEMONSTRATION ON A MOBILE ROBOT

In nature, creatures are capable of actively detecting environmental features and executing corresponding behaviors using contact sensory information obtained through their whiskers under limited visual conditions. To emulate these phenomena, demonstrations of environmental perception are provided by integrating the proposed sensor to a mobile robot. As shown in Fig. 9, a four-wheeled mobile robot is constructed. With two whisker-like sensors whose ends are connected to servos, the robot is able to actively make contacts with the environment through sensor rotation.

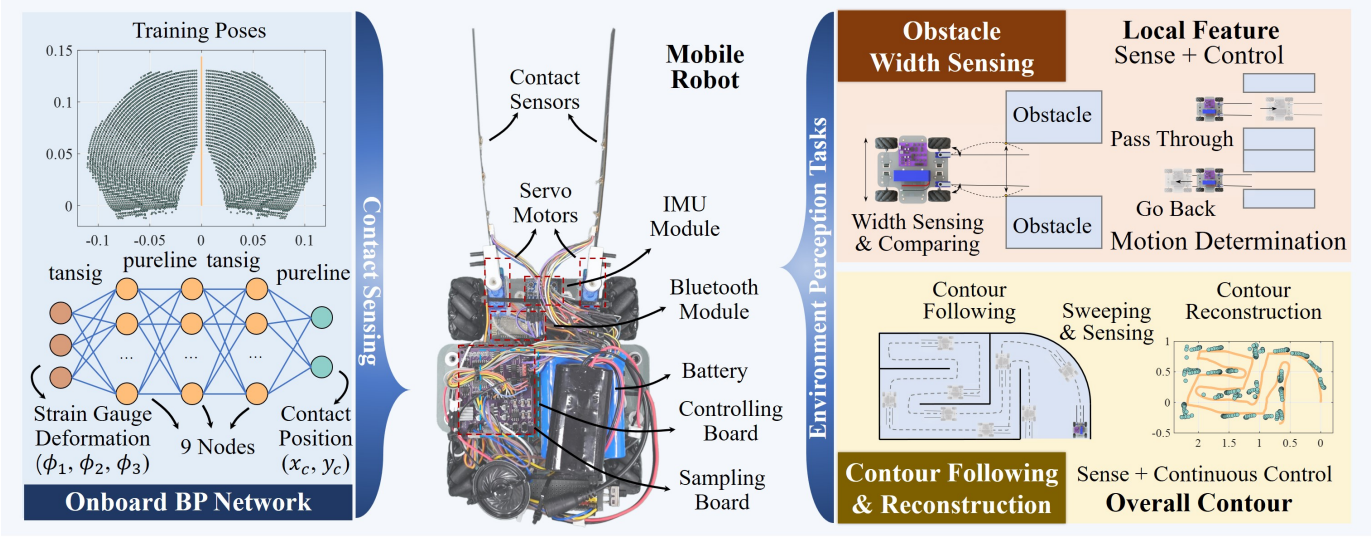


Fig. 9. Environmental Perception Applications of the Proposed Sensors on a Mobile Robot.

Due to the limited computing performance of the microprocessor (Arduino Mega 2560) on the mobile robot, the perception calculation of the demonstration part does not rely on solving the nonlinear equations, but instead utilizes a neural network based on a large amount of model generated data. Using the kinetostatics model in Section II, 5934 sets of contact positions and their corresponding sensor deformation data are quickly generated and used as the input and output data of the neural network training set. A BP network with 3 hidden layers and 9 nodes in each layer is trained using the generated datasets. Leveraging the network's architecture, the computational process of the trained net is transformed into matrix operations through weight and bias matrices. As a result, an efficient contact sensing model that can be used for low-performance processors is established, achieving contact position estimation at frequencies exceeding $90\,Hz$ ($11\,ms$ calculation time) on the microprocessor of the mobile robot while maintaining similar accuracy as the iterative-solved contact sensing model. Considering the signal acquisition system's $100\,Hz$ sampling rate ($10\,ms$ acquisition time) and approximately $8\,ms$ actuation latency in the wheel motor control loop, the mobile robot operates at approximately $35\,Hz$ for its complete sensing-to-action process, which satisfies real-time performance requirements for the demonstrated operational scenarios.

To illustrate the sensor's environmental perception capabilities, the following two scenarios are set up, demonstrating its ability to detect local width information as well as reconstruct large-scale contours in a continuous way.

A. Channel Width Identification

Studies have demonstrated that rats are capable of perceiving the width of narrow apertures using their whiskers [34]. Similarly, in this scenario, the mobile robot equipped with two whisker-like sensors is capable of performing width detection of narrow channels. As shown in Fig. 10, the robot is placed

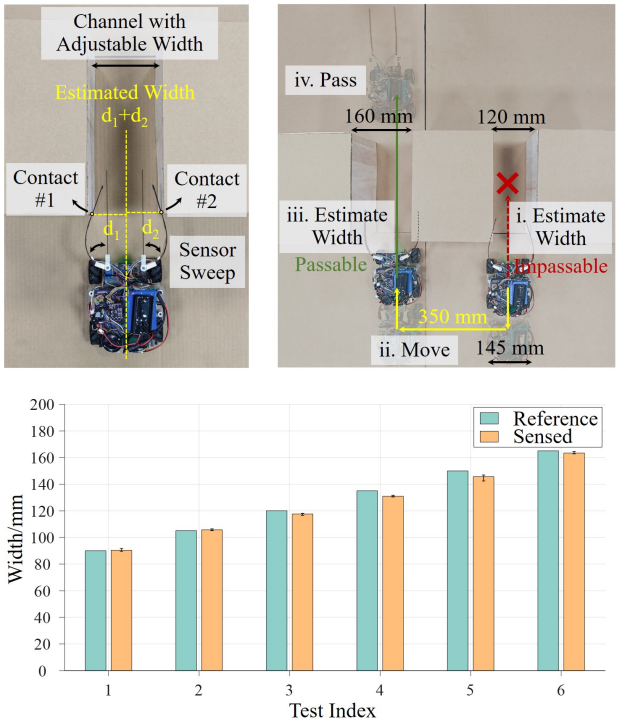


Fig. 10. Result of Channel Width Identification.

so that its center line is in parallel with a narrow channel. By rotating the two whiskers and making contact with both size of the channel, the robot is able to measure the width of the channel from the obtained contact position coordinates. The sensors' rotational range is set from straight forward position to 60° outward direction, with a sweeping cycle completed in 8 seconds. The relatively slow sweeping motion enables the valid assumption of quasi-static contacts, where the proposed sensing model remains applicable. The channel is adjusted to

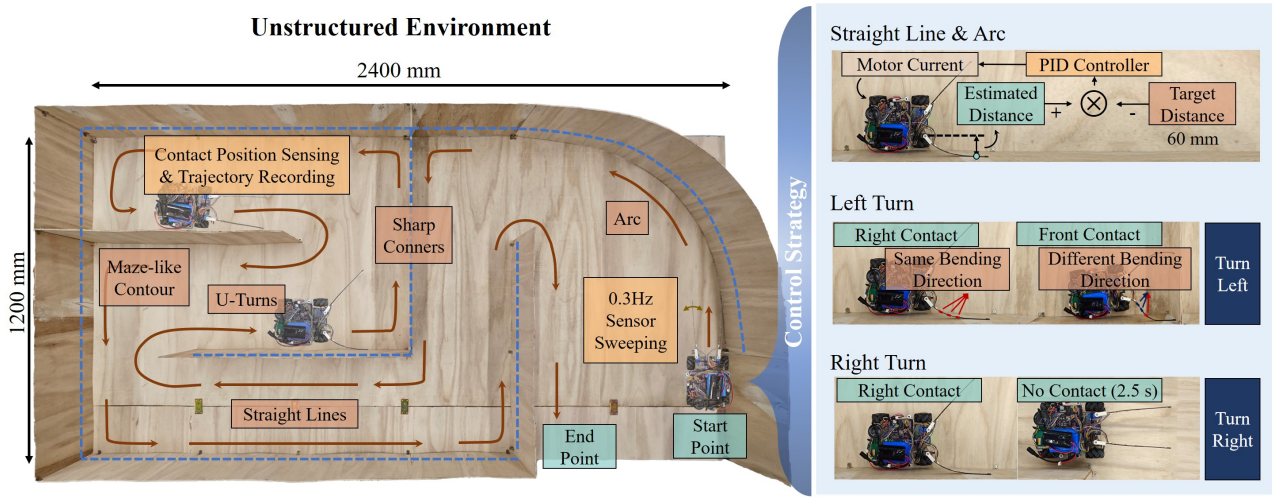


Fig. 11. Unstructured Environmental Perception and Reconstruction of the Mobile Robot Adopting a Right-Wall Following Strategy.

six different widths from $90\,mm$ to $165\,mm$. For each channel width, five measurements are taken. The test results show that the robot is able to sense the width of the channels with an average error of $2.24\,mm$, and the deviations of different trials are all within $4.50\,mm$.

Furthermore, beyond merely sensing the width, the mobile robot is able to evaluate whether it can pass the channel by comparing the perceived width with its own dimensions. A demonstration scenario is constructed in which the mobile robot needs to move from one side of obstacles to the other. As shown in Fig. 10, two parallel channels, with fixed widths of $120\,mm$ and $160\,mm$ respectively, connects the two sides. The robot first senses the narrower channel and concludes it to be impassable. Subsequently, the robot shifts laterally to detect the wider channel, determines it to be sufficiently wide, and proceeds forward through the channel to reach the other side of the obstacles. This process is presented in the supplementary video[1].

B. Contour Following and Reconstruction

In addition to detecting localized features, contour following and reconstruction in large-scale, unstructured environments can be achieved by utilizing the proposed sensor's precise perception capabilities, combined with the robot's mobility. Here, another application scenario is presented, as shown in Fig. 11, a complex contour field is constructed, including straight lines and arcs, as well as maze-like structure with sharp turns of left and right corners. The robot needs to continuously adapt its locomotion solely based on real-time feedback from the contact sensors, moving through the contour from the start point to the end point, while simultaneously recording contact information and reconstructing the contour, all without prior knowledge of the contour environment.

During operation, the robot's whisker sensors sweep at a frequency of 0.3 Hz to detect external contact conditions. The sweeping range for each sensor remains at $60°$. Based on the contact information, the robot executes a wall-following strategy to follow the contour and solve the maze, ensuring that the sensor on the left or right side remains in contact with the contour [35].

Taking right-wall following as an example, for smooth contours like straight line and arc, the robot senses its distance from the right-side contour using the right whisker sensor, calculates the deviation from a target distance ($60\,mm$), and determines the wheel motor speed via a PID controller to maintain a relatively constant distance from the obstacle. Front-facing obstacles are identified based on the deformation direction of the strain gauges on the sensor. When the sensor makes contact with a right-side obstacle, all strain gauges exhibit the same bending direction. However, if a front obstacle is encountered, at least one strain gauge will bend in a direction different from the others during sensor sweeping. Upon detecting this condition, the program commands the robot to execute a left-turn maneuver. Conversely, if the right-side sensor fails to detect contact for an extended period (2.5 seconds), the program interprets this as a sharp rightward change in the contour and commands the robot to perform a right-turn motion until contact is regained on the right side.

During contour-following tasks, the contact points detected by the whisker sensor at each moment are recorded, alongside the wheel encoder measurements of rotational speed and the robot's motion direction as measured by the IMU module. This data is transmitted to an upper computer via the robot's onboard Bluetooth module. By designating the robot's initial position as the origin, the trajectory of the robot's center point and the positions of the contact points relative to the ground coordinate system can be determined and visually presented in real time through a MATLAB program. A real time demonstration of contour following and contact points recording is presented in the supplementary video[2].

The experimental is conducted for multiple times, and the results demonstrate that relying solely on contact information

[1]See supplementary materials - S1.

[2]See supplementary materials - S2.

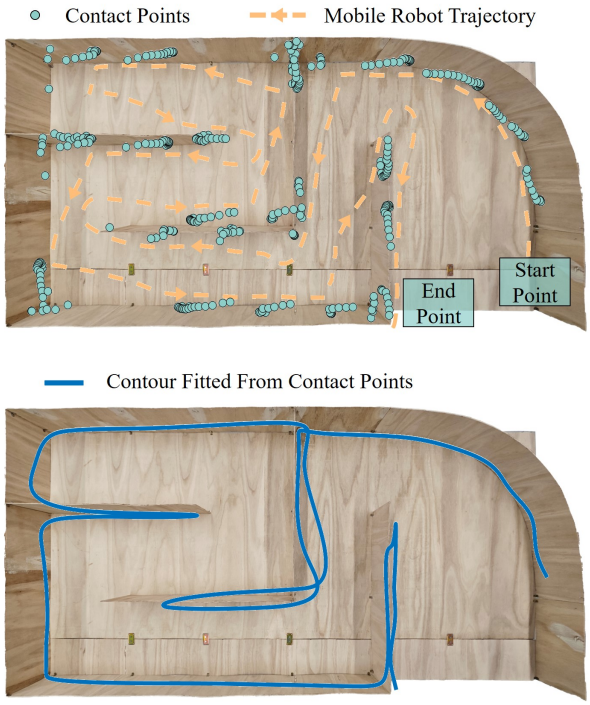


Fig. 12. Result of Contour Following and Reconstruction.

obtained from the proposed contact sensors, the mobile robot can follow straight line and arc-shaped obstacles, execute 90° or 180° turning maneuvers to pass the maze-like contour and reach the end point. A stable contact state can be regained after losing contact with the obstacle due to the mobile robot's limited movement control precision. The results of the trajectory and contact points for one trial are illustrated in Fig. 12. Moreover, by utilizing the contact point data perceived during the following process, the mobile robot is capable of reconstructing the contour of the obstacle it is following. Fig. 12 shows the fitted contour using splines. The fitted contour shape generally aligns with the actual contour. However, reconstruction error caused by the recording accuracy of encoders, IMU module, and the open-loop controlled servo motor angles, can be observed.

VI. Conclusion and Future Work

This paper presents a whisker-like sensor for contact point perception using elastic beams, which can be applied in environmental exploration of robotic systems. The proposed sensor leverages the principles of continuum mechanics, which achieves a simple structure and fabrication process. Through an explicit model that characterizes the force-deflection behavior of the elastic beams, the sensor accurately estimates position of intermediate external contact points along its length, even under non-frictionless and large-scale deformation conditions. Experiments have validated the sensor's capability in detecting contact points over a wide range of test configurations. Furthermore, applications through a mobile robot equipped with the sensors have showcased potential usage in environmental perception, including width sensing, contour following, and reconstruction. Future work will focus on improving the sensor's ability to detect out-of-plane and multi-point contacts, the miniaturization of the utilized strain gauges, and structural design of the sensor for more application scenarios.

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